

Daniel, a patient living with ASCVD + CKD who has experienced recurrent CV events

In patients with prior MI, the presence of inflammation is associated with an increased risk of MACE (28%), all-cause death (42%), and CV death (44%).^{1*}



Daniel, 70.

ASCVD + CKD

2 months post-second MI

Blood Pressure: 132/78 mmHg

LDL-C: 62 mg/dL

eGFR: 37 mL/min/1.73m²

Current Tx: Anti-platelet,
Intensified statin therapy, RAASi,
Beta blocker, SGLT2i

- Daniel quit smoking 8 years ago and he's drinking less; his diet includes lots of salty snacks, and his walks are getting shorter.
- He's on SoC for his conditions and says he hasn't missed a dose. On the surface, Daniels' CV risk seems well managed. His LDL-C is nearly at goal. His blood pressure is steady.
- You remember something you heard at a recent conference: **Inflammation is a greater driver of recurrent CV events than LDL-C in statin-treated patients with ASCVD and CKD.**^{2†}

Inflammatory risk can be detected using the prognostic biomarker hsCRP (≥ 2 mg/L).³⁻⁵

- You check Daniel's hsCRP levels to assess his inflammatory risk. The result comes back:

2.6 mg/L.

- Two months on from his most recent MI, hsCRP 2.6 mg/L is higher than you expected.



- The 2025 ACC Scientific Statement on inflammation in CVD states that hsCRP ≥ 2 mg/L signals an increased risk for recurrent CV events, irrespective of LDL-C levels.⁶
- Daniel's LDL-C is nearly at goal, and his blood pressure is stable, so you focus on lifestyle changes to lower his CV risk.
- You explain that CV inflammation is adding to Daniel's CV risk and suggest he build up his walking distance again and improve his diet by cutting out salty snacks.
- Daniel is surprised, he thought he was on the right track. But he's motivated to turn things around. He agrees to your suggestions and says he will try to improve his lifestyle.

Are you using hsCRP testing to fully stratify inflammatory CV risk in your patients with ASCVD + CKD?



Footnotes:

*A healthcare-based study of 17,464 patients with a median hsCRP level of 2.2 (interquartile range, 1.0–6.0) mg/L and a median of 2.2 (interquartile range, 0.8–4.9) years since their MI. Adjusted for age, sex, time since MI, haemoglobin, eGFR, undertaken procedures, and ongoing medications. Compared with hsCRP <2 mg/L, patients with hsCRP ≥ 2 mg/L had a higher risk of MACE (n=3,900, adjusted HR 1.28, 95% CI, 1.18–1.38), all-cause death (n=4,138, 1.42, 1.31–1.53), and CV death (n=2,343, 1.44, 1.29–1.60).¹

†A CANTOS sub-study evaluating the contributions of residual cholesterol risk (assessed via LDL-C and non-HDL-C) and residual inflammatory risk (assessed by hsCRP and IL-6) as determinants of MACE and total mortality in 9,151 stable statin-treated post-MI patients followed for up to 5 years (median 3.7 years). Results were stratified by baseline eGFR above or below 60 mL/min/1.73 m² (all participants had eGFR >30 mL/min/1.73 m²). In those with eGFR <60 mL/min/1.73m², increasing quartiles of hsCRP significantly predicted recurrent MACE [HR comparing the top to bottom quartile for hsCRP 1.50 (p=0.021)] whereas increasing quartiles of LDL-C did not [HR comparing the top to bottom quartile for LDL-C 1.04 (p=0.80)].²

References:

- 1: Carrero JJ et al. J Am Heart Assoc 2019; 8:e012638.
- 2: Ridker PM, et al. Eur Heart J. 2022; 43(46):4832–4844.
- 3: Ridker PM et al. Lancet 2023; 401(10384):1293–1301.
- 4: Ridker PM. J Am Coll Cardiol 2018; 72:3320–3333.
- 5: Mazhar F, et al. Eur Heart J 2024; 45(44): 4719–4730.
- 6: Mensah GA, et al. J Am Coll Cardiol. 2025; 0(0).

Abbreviations:

ACC=American College of Cardiology, ASCVD=Atherosclerotic cardiovascular disease, CKD=Chronic kidney disease, CV=Cardiovascular, eGFR=Estimated glomerular filtration rate, hsCRP=High-sensitivity C-reactive protein, LDL-C=Low-density lipoprotein cholesterol, MI=Myocardial infarction, Tx=Treatment, SoC=Standard of care.